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Dear Editor,

We submit our manuscript entitled “*Adaptive Multi-Representation Fusion via Dual-Channel PCNN with Multi-Scale Convolution for Vibration Signal Fault Diagnosis*” for consideration in *Signal, Image and Video Processing*.

Single-representation methods for vibration-based fault diagnosis capture only partial signal characteristics, and existing fusion strategies lack adaptivity in regulating heterogeneous representations. We propose a two-stage framework: an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) that fuses STFT spectrograms and GADF images via contrast-guided dynamic weighting, paired with a multi-scale convolution and channel attention enhanced VGG16 (MSCA-VGG16) classifier. The main contributions are:

1. **AW-DPCNN fusion mechanism:** An adaptive fusion mechanism is proposed that integrates heterogeneous STFT spectrograms and GADF images via contrast-guided dynamic weighting and dual-channel convolutional stimuli, adaptively regulating per-channel contributions to suppress destructive interference and enhance feature complementarity.
2. **MSCA-VGG16 classification network:** A dedicated classifier is designed with multi-scale convolution for capturing fault patterns at multiple temporal and spectral granularities, squeeze-and-excitation channel attention for dynamic frequency-band recalibration, and a compact embedding head for enhanced intra-class compactness and inter-class separability.
3. **Comprehensive experimental validation:** Extensive experiments on the CWRU bearing dataset, including backbone and representation comparisons, systematic ablation studies, and noise robustness evaluations, with cross-sensor generalization to fan-end data (97.45%) and cross-sampling-rate robustness at 48 kHz (94.81%).

This work aligns closely with the scope of *Signal, Image and Video Processing*. The manuscript bridges signal and image processing: vibration signals are encoded into complementary image representations (STFT spectrograms and GADF images), and the AW-DPCNN mechanism performs adaptive, contrast-guided fusion of these heterogeneous image modalities—directly addressing the journal’s emphasis on **adaptive processing**, **image processing** (fusion, enhancement, multi-spectral processing), and **multi-resolution signal analysis**. The MSCA-VGG16 classifier further contributes to **non-linear signal processing** and **modeling** through multi-scale convolution and channel-wise attention. Beyond bearing diagnostics, the proposed framework generalizes to heterogeneous image fusion and adaptive representation learning in acoustic monitoring, industrial inspection, and multimedia signal processing, underscoring its broad relevance to the signal and image processing community.

Thank you for your time and consideration.

Sincerely,

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